1 Database management systems

Week 1 · Worksheet

Name _____ Date _____

Exercises

Show your work. Each exercise has a hint at the end: try without it first.

1. Spot the redundancy Warm-up · Lesson 1

A flat file stores: order, customer name, customer phone, product. Ana ordered three times. Which values repeat, and what can go wrong?

Hint: Look at the columns that describe the customer, not the order.

2. Before and after Practice · Lesson 1

List three problems of file-based systems and the DBMS feature that solves each one.

Hint: Think of redundancy, recovery, and security.

3. Name the mini-world Warm-up · Lesson 2

For a school library database, describe its mini-world in one sentence and name two kinds of facts it stores.

Hint: Who borrows what, and when?

4. Design a library Practice · Lesson 6

A library stores members (code, name), books (code, title), and loans (member, book, date). Propose the tables, their primary keys, and the foreign keys in Loan.

Hint: A member can have many loans; a book can be lent many times.

5. Find the hidden N:M Challenge · Lesson 6

A music app stores playlists and songs. A song can be in many playlists, and a playlist has many songs, in a chosen order. Design the junction table.

Hint: Where does "position in the playlist" belong?

6. Write the query Warm-up · Lesson 9

List the name and price of products that cost less than 10,000, cheapest first.

Hint: WHERE for the condition, ORDER BY … ASC for the order.

7. Who bought what? Practice · Lesson 9

Write a query that shows each customer's name next to the id of every order they placed.

Hint: Join customer and orders on customer_id.

8. Name the attribute types Warm-up · Lesson 10

For an Employee: employee_id, full_name (first + last), emails (several), hire_date, years_of_service. Classify each attribute.

Hint: Ask: does it identify, split, repeat, or get calculated?

9. Spot the weak entity Practice · Lesson 10

A loan is paid in instalments numbered 1, 2, 3… for each loan. Which entity is weak, and what is its key?

Hint: Instalment number 2 exists in every loan.

10. Map a course catalogue Practice · Lesson 11

Students enrol in many courses and courses have many students; each enrolment has a grade. Write the tables.

Hint: N:M with an attribute on the relationship.

11. Natural or surrogate? Challenge · Lesson 11

Should a customers table use the email address as its primary key?

Hint: Can a customer change their email?

12. DDL or DML? Warm-up · Lesson 12

Classify: CREATE TABLE, UPDATE, DROP TABLE, INSERT, ALTER TABLE, DELETE.

Hint: Does it change the structure or the rows?

13. Write the table Practice · Lesson 12

Write CREATE TABLE for supplier: an id, a required name, and an email that cannot repeat.

Hint: PRIMARY KEY, NOT NULL, UNIQUE.

Quick check

Circle one answer.

14. What is the main risk of storing the same customer in several files?

A) The copies become inconsistent B) The files load faster C) Programs need less code

15. Which statement is information, not just data?

A) "Customer 2 lives in Lima" B) "Lima" C) "2"

16. A data analyst writes her own SQL queries and small tools to study sales. What kind of end user is she?

A) Sophisticated B) Parametric C) Casual

17. Who is responsible for granting access and monitoring performance?

A) The database administrator (DBA) B) A casual end user C) The application programmer

18. Orders contain many products, and products appear in many orders. What do you add?

A) A junction table with order_id and product_id B) A product_id column in Orders
C) A list of product names in one cell

19. An INSERT into Orders with customer_id = 99 fails with a foreign key error. Why?

A) No customer with id 99 exists B) Order ids must be even C) Foreign keys must be text

20. A sale creates an order, adds the item, and reduces stock. The power fails after step two. What should happen?

A) The DBMS undoes the partial sale using its log B) The order stays without stock change
C) The whole table is deleted

21. Which level describes files, pages, and indexes?

A) Internal B) Conceptual C) External

Answer key

1. Ana's name and phone repeat in all three rows. If her phone changes and only one row is updated, the file contradicts itself.
2. Redundancy → data stored once and shared. Damaged files → backup and recovery with a change log. Anyone can read everything → access control per user and per table.
3. The mini-world is the library: its members, books, and loans. It stores facts such as "member 7 borrowed book 120 on 3 May" and "book 120 has 2 copies".
4. Member(member_id PK, name), Book(book_id PK, title), Loan(loan_id PK, member_id FK → Member, book_id FK → Book, loan_date). Member 1:N Loan and Book 1:N Loan.
5. Playlist_Song(playlist_id FK, song_id FK, position) with primary key (playlist_id, song_id). Position belongs to the pair, not to the song or the playlist alone.
6. SELECT name, price FROM product WHERE price < 10000 ORDER BY price ASC;
7. SELECT c.name, o.order_id FROM customer AS c JOIN orders AS o ON c.customer_id = o.customer_id;
8. employee_id: key. full_name: composite. emails: multivalued. hire_date: simple. years_of_service: derived from hire_date.
9. Instalment is weak. Its key is (loan_id, instalment_no).
10. student(student_id PK, name); course(course_id PK, title); enrolment(student_id FK, course_id FK, grade, PK(student_id, course_id)). The grade lives in the junction table because it belongs to the pair.
11. Better not. Emails change and would have to be updated in every foreign key. Use a surrogate customer_id and declare email UNIQUE.
12. DDL: CREATE TABLE, DROP TABLE, ALTER TABLE. DML: UPDATE, INSERT, DELETE.
13. CREATE TABLE supplier (supplier_id INTEGER PRIMARY KEY, name VARCHAR(100) NOT NULL, email VARCHAR(150) UNIQUE); Try it in the SQL Lab task "Create a table".
14. **A** — When one copy changes and the others do not, the data contradict each other.
15. **A** — It places the values in context. Alone, "Lima" or "2" have no clear meaning.
16. **A** — Sophisticated users know the database well enough to build their own queries and tools.
17. **A** — Access, security, performance, and structural control belong to the DBA.
18. **A** — Many-to-many needs a third table holding both foreign keys, plus details such as quantity.
19. **A** — Referential integrity: the foreign key must match an existing primary key. Create the customer first.
20. **A** — Transactions are all or nothing. Recovery uses the log to roll back incomplete work.
21. **A** — The internal level is physical storage. Conceptual is the logical model; external are user views.